MX17 — Thermal-window ($> 1\,\mathrm{ms}$) Trigger Optimization

SiPM wall $\times$ plastics: tagging X17 & IPC pairs, rejecting capture-γ background

Geant4 full-simulation study

2026-07-19 *(final surveyed geometry)*

The question

Goal: with the final geometry, decide how to trigger in the *thermal neutron* time window (arrival $> 1\,\mathrm{ms}$ after the γ flash) to efficiently tag possible **X17** and **IPC** e^+e^- pairs while rejecting neutron-capture backgrounds.

Two trigger detectors per arm

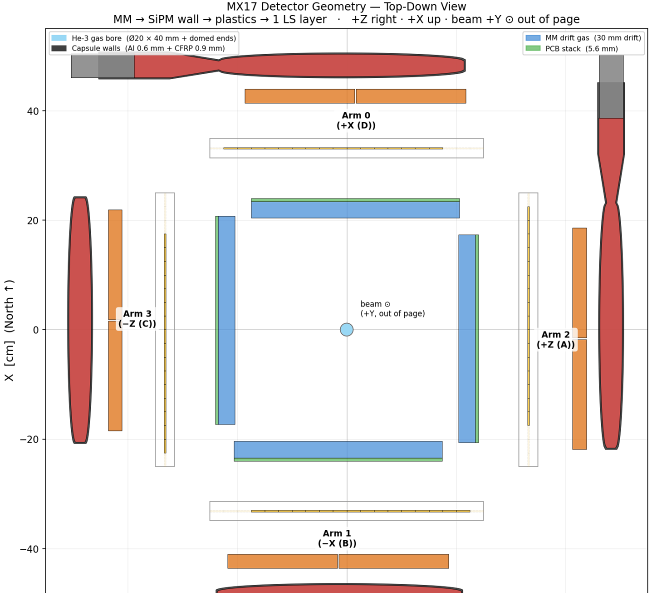
- SiPM wall — 16 thin ($3\,\mathrm{mm}$) scint bars; gives the 2-arm pair *topology*.
- Plastics — 2 thick ($2.5\,\mathrm{cm}$) bars; give *energy* (per-leg pulse height).

Design levers

- SiPM per-bar threshold a
- Plastic per-bar threshold b
- Coincidence topology (1 leg / 2 legs / ...)

Both thresholds are quoted in **MIP** units, so they transfer directly to hardware via the measured MIP peaks. $1\,\mathrm{MIP} = 458\,\mathrm{keV}$ (SiPM bar), $4.33\,\mathrm{MeV}$ (plastic).

Setup — 4-arm detector, top-down



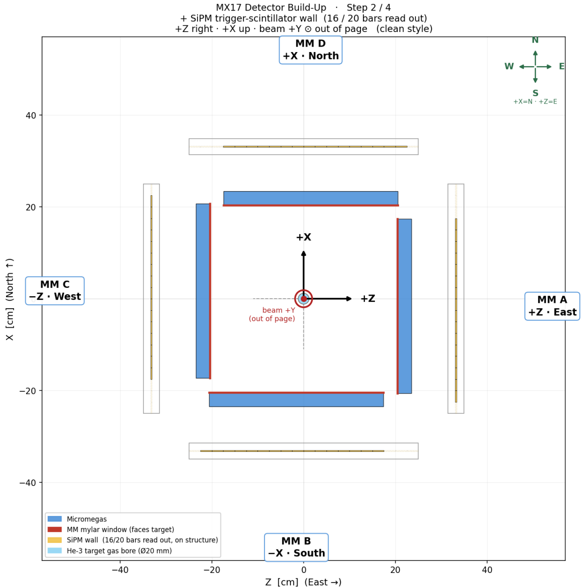
Beam $(+Y)$ into the page; four arms in the transverse plane.

Stack, inside $\to$ out per arm:

1. Micromegas (tracking)
2. **SiPM wall** (trigger)
3. **plastics** (trigger)
4. liquid-scint layer

He-3 target at the centre: $\varnothing 20 \times 40$ mm gas bore in an Al+CFRP capsule.

Setup — the trigger stack



SiPM wall (yellow): 50 cm span, 16 of 20 bars read out, bars run along the beam, array along the tangent. Centered on the mechanical structure.

Plastics (orange): two $20 \times 30 \times 2.5$ cm bars, tape+Al wrapped, behind the wall, centered on the Micromegas.

A leg "fires" when its SiPM bar *and* a plastic bar are both above threshold.

Simulation

Campaigns (Geant4 11.2, FTFP_BERT_HP; final geometry; all files validated):

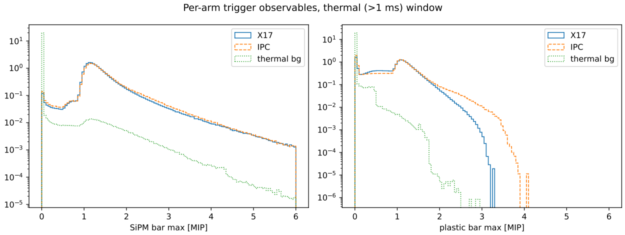
Run	primaries	purpose
thermal n	10^9	in-gate bkg
	1 meV–2 eV	
epi n	5×10^8	spill-in
	2 eV–100 keV	check
pairs	10^7	X17+IPC
	50/50	signal

Two physics details that matter

- **Self-shielding:** thermal captures sit within $\sim$mm of the gas entrance (median 14 mm depth, 95% $<$ 25 mm). Pair vertices are sampled from the *measured* capture profile, not uniform-in-gas ($\sim$3.7 cm upstream of centre).
- **Gate $\leftrightarrow$ energy:** $\mathrm{TOF[ms]} = 1.41/\sqrt{E_n[\mathrm{eV}]}$ at 19.5 m; $> 1\,\mathrm{ms} \Leftrightarrow E_n < 2\,\mathrm{eV}$, 4.28×10^6 n/pulse.

Geant4 has no internal pair conversion — X17 and IPC are explicit generator modes; the neutron runs supply the capture-γ background and all singles.

Trigger observables — per-arm pulse height in MIP units



Signal legs peak at 1 MIP (crossing $e^{\pm}$); stopped legs reach ∼3 MIP in the plastic.

Background (thermal captures):

- plastic hits die above ∼1.7 MIP — the ^{28}Al 7.72 MeV Compton edge; H-capture 2.22 MeV γ stays < 0.5 MIP.
- SiPM shows no MIP peak (γ Comptons only).

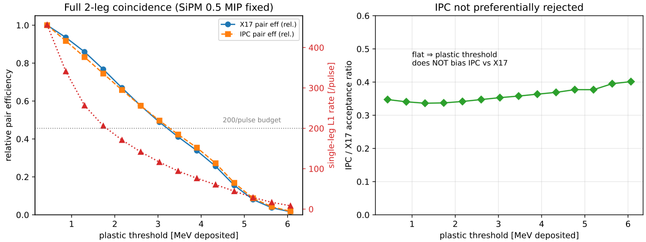
$\Rightarrow$ the **plastic** pulse height is the background-rejection lever.

Topology: full coincidence costs efficiency, but is clean

Pair trigger (both arms)	ε(X17)	ε(IPC)	bkg pair-tags/pulse
SiPM-only (2 arms ≥ 0.5)	15.0%	—	42
2 SiPM $+ \geq 1$ plastic confirm	7.0%	5.3%	1.8
full: 2×(SiPM×plastic)	**2.1%**	**0.7%**	**0.17**

- Requiring the plastic on *both* legs costs $\sim 5\times$ in signal — pure solid angle (plastics are smaller and further back).
- **This study assumes the full coincidence** (the clean, high-purity choice), and optimizes the plastic threshold b within it.

Key result: the plastic threshold does *not* reject IPC



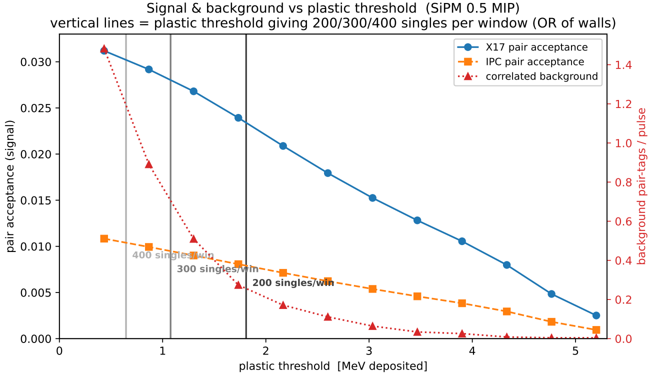
IPC/X17 acceptance ratio is **flat** (∼0.35) across the whole threshold range.

Why: IPC's $1/M_{ee}$ spectrum favours soft, collinear pairs — 97% of triggering IPC has both legs in the *same* arm and is lost to the 2-arm requirement *on geometry*, before any energy cut.

The surviving IPC looks kinematically like X17, so *b* cuts them in lockstep.

You may raise the plastic threshold freely for background/rate control: it costs total statistics, **not** the IPC-vs-X17 balance.

Signal-to-noise optimization



Signal (left, blue/orange) falls *gently*; the correlated capture background (right, red) **collapses by $\sim$1.5 MeV** once you clear the capture-γ Compton edges.

So S/N keeps improving with threshold — the cost is total statistics, not the IPC/X17 balance.

Vertical lines: the plastic threshold at which the (simulated) singles rate — the **OR of all four walls** — equals 200 / 300 / 400 per window, i.e. the DAQ capacity ceilings.

Sweet spot **plastic $\approx$ 2–3 MeV**: background already $<$ 0.1/pulse, $\sim$50–65% of X17 & IPC kept, IPC/X17 undistorted, and the singles rate is under the 200/window DAQ ceiling.

Rate budget & the S/N table

plastic [MeV]	ε(X17) pair	ε(IPC) pair	corr. bkg /pulse	single-leg rate/pulse	
0.9	0.029	0.010	0.89	341	
1.7	0.024	0.008	0.27	206	$\leftarrow$ 200 budget
2.6	0.018	0.006	0.11	142	**recommended**
3.0	0.015	0.005	0.06	116	
3.5	0.013	0.005	0.03	94	MC floor

- **Measured DAQ capacity** (`zs_ipd_safety`, 2026-07-18): ∼101 ev/window at IPD=10 (clean), ∼189 at IPD=5, ∼226 at IPD=2 — so the **200/window budget is real** (aggressive today; 300–400 awaits the network upgrade).
- Single-leg (L1) singles are the DAQ constraint — the *pair* coincidence is < 2/pulse. If the 2-arm coincidence is formed in hardware, b can go as low as you like.
- **Singles caveat:** the vertical lines use *simulated* singles (∼450–550/window at the floor), $\sim 2\times$ below the DAQ-measured trigger level — so the true 200/300/400 thresholds sit somewhat higher. Re-anchor once a clean scaler-based singles-vs-threshold scan exists (M5 scaler was dead in the July run).

Summary & recommendation

- **Trigger:** full 2-arm SiPM$\times$plastic coincidence, SiPM ≥ 0.5 MIP, **plastic $\approx$ 2.6–3.0 MeV (0.6–0.7 MIP)**.
- **Keeps both signals:** the plastic threshold does not bias IPC vs X17 (flat ratio); it trades total statistics for purity.
- **S/N-optimal in the measurable range**, correlated background < 0.1/pulse, comfortably under the single-leg rate budget.

Caveats / next steps

- Thermal-gate signal production is tiny (self-shielding ceiling $\sim 10^{-4}$ IPC/pulse) — this window is background characterization; the physics ROI is the μs-scale MeV window (rates $\times 10^4$): re-run the same scan there.
- Accidental (uncorrelated) coincidences scale as (single-leg rate)^2; full optimization needs the coincidence window — Python time-domain layer.
- If the *IPC invariant-mass spectrum* is a goal, keep b lower and correct with efficiency (a high b sculpts the accepted M_{ee} shape).